Degly Sebastián Pava Pava

(+33) 0767880906 (+1)6306355612 Graduation Date: June 2023

dspavap@unal.edu.co https://deglypava.com IMLEX and Systems and Computer Engineer

From: August 2020
To: June 2023
(Currently in France)

Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- *University of Eastern Finland, Toyohashi University of Technology (Japan), University Jean Monnet Saint-Etienne (France) and KU Leuven (Belgium)*

Photonics | Physical Optics | Realtime 3DRendering | Computational Imaging | CUDA | Computer Graphics | Machine Learning | VR/AR.

WORK EXPERIENCE

From: January 2018
To: August 2020

Lead XR Developer | Hackspace for Innovation and Visualization in Education

Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)

- Part of THE HIVE Laboratory working in The HoloBrain and MR Manikin / OCS projects. In charge of the creation of multiple 3D Experiences and their integration into a Mixed Reality environment while using concepts of 3D visual computing, computational geometry and VR, AR and MR toolkits within the Unity 3D Engine.
- HoloBrain project in collaboration with the Microsoft Garage Internship programme. Developed a volumetric rendering of an MRI scanned brain efficient enough to work on HoloLens v1.

Tech: Unity | Microsoft HoloLens 1&2 | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy | Azure

From: February 2017
To: September 2019

Lead Developer and Engineer at Pontifical Xaverian University

Department of Industrial Engineering and Systems Engineering at Pontifical Xaverian University, Bogota (Colombia)

- By using the NetLogo programming language I developed a simulation of a production planning, scheduling, and inventory control system used to manage manufacturing processes, also known as a Material Requirements Planning. It is currently being used in production Classes to teach new students about MRP and scheduling.

Tech: NetLogo | Anylogic | Microsoft Office Suite | Adobe Suite | Research Techniques

EDUCATION AND TRAINING

Master's studies:

IMLEX + MANUTECH Sleight Scholarship + ERASMUS Mundus Japan Scholarship

Bachelor studies:

Systems and Computer Engineer. Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia.

Trained in multiple research groups:

HIVE Research Lab – ELAP (Emerging Leaders of the Americas Program) Scholarship

The University of British Columbia, Vancouver, B.C. (Canada)

From: January 2019
To: April 2019

- Emerging Leaders of the Americas Program Scholarship winner offered by the Canadian Government.
- Visiting International Research Scholar Fellowship.
- Developed neuroanatomy-based applications while making use of MR and VR technologies for educational purposes.

From: June 2018
To: December 2018

Gold IronHacks Purdue-UNAL 2018 Fellowship – RCODI

Purdue University, West Lafayette, Indiana (United States)

- Participated on the team that organized the Gold IronHacks Purdue-UNAL Hackathon 2018.
- Created and managed parts of the website where the hack would be held.
- Part of the judges that would choose the best implementation of Open Data within a participant created website.

PERSONAL SKILLS

Spanish	Mother Tongue	**French**	Basic Writing and Speaking Skills
English	IELTS Academic (8.0) and iBT TOEFL (103 points)	**Japanese**	Hiragana / Katakana Basic

Digital skills	Extended Realities	Programming	Optics / Physics	Tool and Services
	Unity | Blender | ThreeJS	C# | Python	Photonics /	ImageJ3D | Medical Software
	AWS | Azure	Javascript | Netlogo	Color Science	TensorFlow / TensorBoard
	Vuforia | MRTK | Arduino	Git (GitHub) / SCRUM	Spectral Imaging	Adobe Suite

AWARDS AND PROJECTS

- ***HoloBrain 2.0 – ELAP Scholarship - THE HIVE at The University of British Columbia***
  A mixed reality brain simulation to allow educators to teach the anatomy of the brain with a holographic model, using 3D reconstructions of basal ganglia nuclei which were obtained from MRI scans. Developed a volumetric rendering that could behave as a 3D Mesh and be interactable with hand gestures so the user could cut the brain in real time to have a better idea of its parts and how it was internally organized.

- ***MR. Manikin / Online Clinical Simulation – VIRS Fellowship - THE HIVE at The University of British Columbia***
  A mobile AR application that would enable medical students to interact with a critical care patient. Developed a 3D model of the patient, used Vuforia Engine to place it on the real world and Photon Networking within Unity 3D to send instructions from the Teacher Facilitator (iPad or PC) to the student interacting with the patient (Android or iOS). The application is currently in beta, soon to be published on the App Stores.

- ***Energy Modelling Tool – RCODI's Fellowship - Purdue's Research Center for Open Digital Innovation***
  HTML and AWS based software focused on helping civil engineers and architects find the best way to build an energy efficient building by using data obtained from EnergyPlus processing, Open Data and MATLAB Cloud Computing for costs calculation.

- ***Winner of Purdue-UNAL Gold IronHacks 2017***
  Developed a website with a mashup that utilized Open Data to help students rent houses near the University of Illinois, Chicago. Open Data was used to create interactive visualizations to give insights and support optimal decisions. The project was developed through Git and judged by a team of experts from Purdue University.

- ***Lead developer, Simulation of TransMilenio – Full Scholarship at Universidad Nacional de Colombia***
  Developed a NetLogo simulation of the Bogota's transport service TransMilenio to try to optimize how buses should be deployed to improve the quality of service. The software simulated how the system works and proposes an algorithm to improve its quality.